Urdu Named Entity Recognition: Corpus Generation and Deep Learning Applications

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Named Entity Recognition (NER) plays a pivotal role in various natural language processing tasks, such as machine translation and automatic question-answering systems. Recognizing the importance of NER, a plethora of NER techniques for Western and Asian languages have been developed. However, despite having over 490 million Urdu language speakers worldwide, NER resources for Urdu are either non-existent or inadequate. To fill this gap, this article makes four key contributions. First, we have developed the largest Urdu NER corpus, which contains 926,776 tokens and 99,718 carefully annotated NEs. The developed corpus has at least doubled the number of manually tagged NEs as compared to any of the existing Urdu NER corpora. Second, we have generated six new word embeddings using three different techniques, fastText, Word2vec, and Glove, on two corpora of Urdu text. These are the only publicly available embeddings for the Urdu language, besides the recently released Urdu word embeddings by Facebook. Third, we have pioneered in the application of deep learning techniques, NN and RNN, for Urdu named entity recognition. Finally, we have performed 10-folds of 32 different experiments using the combinations of a traditional supervised learning and deep learning techniques, seven types of word embeddings, and two different Urdu NER datasets. Based on the analysis of the results, several valuable insights are provided about the effectiveness of deep learning techniques, the impact of word embeddings, and variations of datasets.

CCS Concepts: • **Computer systems organization** → **Embedded systems**; *Redundancy*; Robotics; • **Networks** → Network reliability;

Additional Key Words and Phrases: Resource poor languages, deep learning, Urdu NER corpus, Word2vec, fastText, word embeddings

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1 INTRODUCTION

A Named Entity (NE) refers to a conceptual or real-world object of interest that can be represented by a proper noun. Examples of NEs are Harry Potter, Barack Obama, London, Harvard University, and the like. Named Entity Recognition (NER) plays a pivotal role in a wide range of

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natural language processing tasks, such as machine translation, question-answering, and information retrieval systems [1]. There are two interrelated objectives of NER techniques: to ascertain whether a given word in a sentence is a named entity, and subsequently classify each named entity into an NE-type, such as person, location, or organization [2]. Recognizing the importance of NER, a plethora of NER techniques have been developed for the Western languages, especially for the English language [3]. The effectiveness of these NER techniques for the Western languages have been demonstrated in various real-world applications, such as Apple's Siri and Amazon's Alexa [4].

Urdu is the national language of Pakistan, a country of over 200 million inhabitants and is recognized as an official language in multiple Indian states [5]. Urdu speakers can also be found abundantly in UK, USA, Canada, and Norway [6], making the total number of Urdu speakers over 490 million worldwide [7]. Furthermore, Urdu and Hindi share a significant portion of their vocabularies, making both languages mutually intelligible [8]. Thus, millions of native Hindi speakers can also communicate in Urdu. A recent study highlighted the increased digitization of Urdu on the web [6]. Although the initial research activities for the Urdu language processing started back in 1980s [9], Urdu NER applications are still inadequate, primarily due to the following challenges:

- *Lack of Attention.* The initial era of Internet was predominated by the Western languages' contents. Therefore, these languages have been the main focus of the research community. Furthermore, the already belated research activities for the Urdu language has been obstructed by the unique challenges posed by the inherent characteristics of Urdu.
- *Lack of Large NE Corpus.* Few Urdu NE corpora have been developed, but either these corpora are not publicly available or not as large as their Western counterparts, hence, thwarting the development and evaluation of Urdu NER techniques.
- *Scarcity of Word Embeddings.* The latest breakthrough in the accuracy of various English language tasks is attributed to word embeddings. For the Urdu language, no word embeddings are publicly available, except the recently released embeddings by Facebook [10].

1.1 Contributions

A recent extensive literature survey has highlighted the scarcity of resources for the Urdu language processing [6]. In particular, there are merely six Urdu NER corpora. These corpora have at least one of the following limitations compared to their Western counterparts: not publicly available, small number of tokens, or fewer NEs. To this end, the main contributions of this article are as follows:

(1) We have developed the largest, diverse, and, a carefully annotated Urdu NER corpus, named MK-PUCIT.[1] The corpus is composed of 926,776 tokens, including 99,718 NEs, which is substantially greater than the existing Urdu NER corpora.
(2) We have generated and released six word embeddings using two state-of-the-art techniques over MK-PUCIT corpus and another massively large news corpus of 28,006,880 tokens. These are the only word embeddings that are freely available for the Urdu language other than the Facebook released word embeddings.
(3) Using all the available Urdu word embeddings, we are the first to apply two deep learning techniques, Neural Network (NN) and Recurrent Neural Network (RNN), for Urdu NER. Furthermore, we have compared these deep learning techniques with two traditional supervised learning techniques, i.e., Conditional Random Field (CRF), and Maximum Entropy Markov Model (MEMM).

[1] MK-PUCIT can be downloaded from https://www.dropbox.com/sh/1ivw7ykm2tugg94/AAB9t5wnN7FynESpo7TjJW8la.

(4) Finally, we have compared the performance of the newly developed MK-PUCIT corpus with the only publicly available Urdu NER corpus, named IJCNLP [11].

This article is significantly different from our previous work [7] in terms of the number of embeddings, supervised learning techniques, datasets, and the size of the developed corpus. Specifically, the key differences are as follows: First, the number of NEs in the MK-PUCIT are significantly higher than the previously developed KPU-NE corpus [7], i.e., 99,718 as compared to 44,480 NEs. Second, compared to a single word embedding used previously, this article uses seven types of word embeddings, including the Facebook released embeddings. Third, we have performed 10-folds of 32 different experiments using the two corpora, seven types of embeddings, two deep learning techniques ($2 \times 7 \times 2 = 28$), as well as experimented with two traditional supervised learning techniques using the two corpora ($2 \times 2 = 4$). Thus, our developed corpus is not only double in size but the experiments are significantly thorough and diverse as compared to the previous study.

The rest of this article is organized as follows: Section 2 highlights the challenges of Urdu NER. Section 3 describes the details of our developed Urdu NER corpus and compares its specification with the existing corpora. Section 4 provides an overview of the newly generated word embeddings for Urdu. Sections 5 and 6 elaborates the experimental setup and analyze the results, respectively. Finally, conclusions are drawn in Section 7.

2 NAMED ENTITY RECOGNITION IN URDU

In this section, we summarize the challenges of Urdu language processing, Urdu NER, and limitations of the existing Urdu NER datasets.

2.1 Challenges of Urdu

Urdu is a computationally complex language as its morphology and syntactic structure stems from various other languages, such as Persian, Sanskrit, English, Turkish, and Arabic [12]. For instance, 'جاہل jahil' (illiterate) and 'جنت junat' (heaven) are borrowed from Arabic; 'آرزو arzoo' (desire) and 'کشور kishwar' (young), are borrowed from Turkish; whereas 'انناس Ananas' (Pineapple) and 'الماری almari' (cupboard) are derived from Portuguese. Furthermore, Urdu is a free word ordering language thus, even if the ordering of the words is changed, their meanings remain the same [13]. For instance, 'خرم نے کھانا کھایا Khurram nay khana khaya' (Khurram has eaten food) can also be reordered as 'کھانا خرم نے کھایا Khana Khurram nay khaya' without changing the meaning of the sentence.

A key characteristic of Urdu language is its context sensitivity both in terms of writing and comprehension. In writing, the shape of a letter may change due to its neighboring letters as well as its position in a word. For instance, in the word 'باب bab' (alphabetically ب ا ب) the letter ب appears twice, but its shape changes at the beginning of the word only. Similarly, Urdu language is also highly context sensitive in terms of comprehension. That is, there are numerous words whose meanings depend upon the surrounding words. For example, in the sentences, 'وہ سحر ہے woh sehar hai' (She is Sehar) and 'سحر کا وقت ہے sehar ka waqa hai' (It is dawn), the word سحر refers to a person and a specific time, respectively.

Additionally, spelling variations, changing alphabets' shapes due to inclusion of a space, and ambiguities in suffixes, further contributes to the challenges of Urdu language processing [6, 13]. For instance, لاہور (Lahore) and لاھور (Lahore) are spelling variations of the same word where a single letter is changed. Furthermore, the inclusion of space in a word changes its depiction while preserving its meaning. For instance, 'کیطرف ki taraf' and 'کی طرف ki taraf' both mean "towards" but an additional space has drastically changed the word's depiction.

2.2 Challenges of Urdu NER

Identifying and classifying NEs from English text has benefited a great deal from capitalization, accurate POS tagging, and availability of huge gazetteers [6, 14]. Additionally, the availability of quality word tokenizer and strong grammatical principles have played a substantial role in English NER techniques. On the contrary, named entity recognition in Urdu language is a challenging task due to the following key reasons [7, 15, 16]: (a) absence of capitalization, (b) context sensitivity, and (c) highly inflectional and agglutinative nature. A brief discussion of these challenges is as follows:

Absence of Capitalization. In the English language, a key indicator of an NE in a sentence is capitalization. However, due to the absence of capitalization such an indicator is not available in the Urdu language, making NEs identification a challenging task.

Context Sensitivity. A token that refers to an NE in a sentence may change its NE type, or the token may not even remain an NE due to change in its context. For instance, in a sentence 'ریاض وہاں گیا Riaz wahan gaya' (Riaz went there), the token ریاض refers to NE type person, whereas in the sentence 'ریاض دارلحکومت ہے Riaz darulhakoomat hai'(Riaz is the capital city), the NE type of the same token is location.

Inflectional and Agglutinative Nature. In Urdu, a token may either change its NE type or even may not remain an NE when agglutinated with other words, making named entity recognition and classification a challenging task. For instance, 'مختار Mukhtar' (a person name) does not remain an NE when combined with 'خود khudh' (self) to form 'خودمختار khudh mukhtar' (autonomous).

Existing Urdu NER Datasets. Our comprehensive search led to the identification of only six Urdu NER datasets. The datasets are: CRL [6], BBC-Urdu [17], EMILLE [18], IJCNLP [9], CRULP [6], and KPU-NE dataset [7], containing 2819, –, 30027, 2817, 4292, and 44480 NEs, respectively (for detailed specifications, see Table 4). The key limitations of these datasets are that either they are not public or have a small number of NEs. Furthermore, for a majority of the corpora, the guidelines used for annotation, the annotation procedure, and the confusion matrix of the annotators are not available, consequently limiting the reliability of these datasets. In particular, CRULP and CRL corpora are currently unavailable from their declared sources [6]. BBC-Urdu and KPU-NE are not publicly available. IJCNLP contains merely 2,817 NEs, whereas the EMILLE corpus contains 200,000 tokens but has a small tokens-to-NEs ratio [6]. Furthermore, the annotation guidelines and the confusion matrix are not available for BBC-Urdu, EMILLE, IJCNLP, and CRULP datasets.

3 MK-PUCIT CORPUS

This section presents our key contribution, the development of the largest Urdu NER corpus, called *MK-PUCIT*. To establish the strength of this contribution, we have compared the specifications of our corpus with all the publicly available NER copora for Urdu language as well as with the corpora of several Western languages. Our newly developed NER corpus is a significant extension of our previously developed corpus, which was not publicly released. Our extended corpus for an under-resource Urdu language is even significantly larger than the well-established NER corpora for resource-rich Western languages. Therefore, we confidently contend that MK-PUCIT offers abundant opportunities to adequately train deep learning techniques. In the following subsections, we present the details of our corpus development.

3.1 Data Acquisition

In order to develop MK-PUCIT corpus, the initial data was taken from our previously developed KPU-NE corpus. Subsequently, we have added 273,924 additional Urdu tokens by scrapping text

Table 1. Recognition Guidelines

NE	Guideline	Examples
PER	A name, nickname, or alias of a human, or a fictional character should be annotated as PER.	بارک اباما (Barack Obama), علامہ اقبال (Allama Iqbal).
	The name of a cast, clan, or family should be annotated as PER.	ملک (Malik), and کھوکھر (Khokhar)
	Titles, relation names, pronouns, reflexive pronouns, name prefixes, and Gods names should *not* be marked as PER.	مسٹر (Mr.), صدر (President), تم (You), جونیئر (Junior/ Jr.)
ORG	The name of a company, media group, team, political party or any other entity created by a group of people should be marked as ORG.	ہارورڈ یونیورسٹی (Harvard University), and میکروسافٹ (Microsoft)
	The name of a product or a brand should *not* be marked as ORG.	سامسنگ گلیکسی نوٹ (Samsung Galaxy Note)
LOC	All man-made structures and politically defined places like the names of countries, rivers, and railway stations are marked as LOC.	لندن (London), نیوزی لینڈ (New Zealand), and پنجاب (Punjab)
	A generic reference to a location or a nationality shouldn't be marked LOC.	نیا ایئرپوٹ (new airport), and امریکن (American).
O	All the remaining words, such as prepositions, adjectives, adverbs, and the names of books, movies, etc. are marked as O.	میں (I), تم (you), وہ (he), ہم (we)

from NewsLink.pk over the period of two years, 2016 – 2017. It is an online portal that collects news from majority of the popular Urdu news websites. The news text was chosen as a seed due to two reasons, (a) news are widely acknowledged as a rich source of NEs, and (b) a large majority of the NE corpora for Western languages are based on news text [19, 20]. To acquire text from Newslink.pk, we developed a program that crawls the website, extracts the Urdu text from html, and stores the extracted text in a relational database. The resultant MK-PUCIT corpus is composed of 926,776 tokens from over a dozen genres, including education, entertainment, politics, sports, and religion.

3.2 Annotation Elements and Guidelines

There is no standard list of NE types that is mandatory for the development of an NER corpus. For the development of our MK-PUCIT corpus, we focused on three NE types: Person (PER), Location (LOC), and Organization (ORG). There were two key reasons for choosing these particular NE types: (a) these NE types have wider application areas, and (b) the most established NER corpora for the Western languages rely on these NE types [7, 19, 21]. We have developed a set of clearly defined guidelines to annotate Urdu text unambiguously. In particular, we have developed two-level guidelines, namely Recognition and Disambiguation guidelines.

Recognition guidelines are a set of rules to identify individual NEs from a given text. That is, the recognition guidelines are applicable when either a sentence contains a single NE or multiple NEs that are separated by other text. Our developed recognition guidelines are presented in Table 1.

Disambiguation guidelines are a set of rules to disambiguate multi-name expressions. A multi-name expression refers to two or more NEs that appear next to each other in a sentence. We have identified two cases that demand disambiguation: Overlapping and Consecutive. *Overlapping* refers to the case when two NEs overlap with each other in a sentence. For example, the sentence 'اس نے شمالی اور جنوبی کوریا کا دورہ کیا os nay shumali aur janoobi koreeya ka doora kiya' (He toured South and North Korea) contains two overlapping NEs, North Korea and South Korea. In such a case, the overlapping NEs should be annotated as a single NE because the two NEs are inseparable. *Consecutive* refers to the case when two NEs follow each other. For example, the sentence 'اس نے شمالی کوریا اور جنوبی کوریا کا دورہ کیا os nay shumali koreeya aur janoobi koreeya ka doora kiya' (He toured North Korea and South Korea) contains two separate but consecutive NEs. In such a case, the NEs should be annotated individually because the two NEs are clearly separable.

Table 2. Confusion Matrix of Kappa Statistic

	PER	LOC	ORG	O	Total
PER	46	6	13	12	77
LOC	6	62	10	15	93
ORG	7	15	33	11	66
O	8	4	3	1124	1139
Total	67	87	59	1162	1375

Table 3. Specification of MK-PUCIT Corpus

Sr. No.	Characteristic	KPU-NE	Extension	MK-PUCIT
1	Total number of tokens	652,852	273,924	926,776
2	Person (PER)	18,150	18,036	36,186
3	Location (LOC)	18,728	24,322	43,050
4	Organization (ORG)	7,602	12,880	20,482
5	Total NEs	44,480	55,238	99,718

3.3 Annotating the Corpus

For annotating the corpus, the two annotators were asked to verify whether a token represents an NE or not. This decision should be based on the context of the token. In case the token was not NE, it was marked as O; otherwise, the category of NE was determined to mark it PER, ORG, or LOC using the Inside-Outside (IO) tagging scheme.

To evaluate the effectiveness of our developed guidelines and the annotation process discussed above, we used two annotators. The annotators were native speakers of Urdu language and have a post-graduate degree in computer science. The annotators were asked to comprehend the annotation guidelines, and annotate a small sample of 100 randomly selected sentences to minimize the human bias. In addition to the guidelines, an explanation and discussion session was organized to develop a clear understanding of the annotation task. Subsequently, the results of the two annotators were compared and inter-annotator agreement was computed using Kappa coefficient [22]. The value of Kappa coefficient for NE identification was found to be greater than 0.7, indicating a very high similarity between the annotators. The confusion matrix between the two annotators is presented in Table 2.

3.4 Specifications and Analysis of MK-PUCIT

Table 3 provides the specification of the seed corpus and our extensions. From the table, it can be observed that we have made substantial extension to the seed corpus. That is, we have annotated 273,924 additional tokens and 55,238 additional NEs. Thus, the total number of NEs in the newly developed corpus are doubled as compared to its seed corpus. Hence, the extended MK-PUCIT corpus contains 926,776 tokens and 99,718 NEs.

To establish the superiority of the newly developed MK-PUCIT corpus, we have compared MK-PUCIT specifications with several existing corpora of Urdu and Western languages in Table 4. Given that, we have annotated three NE types in our corpus; therefore, the comparison with the existing corpora also includes the specifications of these three NE types. In the table, the "-" sign is used when the information is not available. From the table, it can be observed that MK-PUCIT is the largest Urdu NER corpus with 99,718 NEs and it is even larger than the widely used NER corpora of the Western languages.

Table 4. Comparison of NER Corpora

Corpus Name	Year	Language	Tokens	NE types	PER	LOC	ORG	Total NEs
Reuters-21578 [23]	1999	English	33,413	4	269	176	56	>501
BBC-Urdu [17]	2002	Urdu	50,000	–	–	–	–	2,819
EMILLE [18]	2003	Urdu	200,000	–	–	–	–	–
CoNNL-03 [19]	2003	Lang. Indep.	35,089	0	10,059	10,645	9,323	30,027
IJCNLP [6]	2008	Urdu	58,252	12	641	844	122	2,817
GerNED [24]	2012	German	2,648	4	700	563	1,127	2,390
NoSta-D [25]	2014	German	41,005	12	10,500	12,165	7,175	20,390
CRL [6]	2008	Urdu	55,000	–	1,772	1,262	1,258	4,292
MUC7 [21]	1998	English	–	3	336	609	1,042	1,987
OntoNotes NW [26]	2009	English	–	3	27,354	32,282	29,968	89,604
Europeana [27]	2016	French	207,000	3	5,672	5,614	2,574	13,860
	2016	Dutch	182,483	0	4,492	4,448	1,160	10,100
	2016	German	96,735	0	7,914	6,143	2,784	16,841
BIO-NER [20]	–	English	201,166	5	11,692	8,431	9,736	29,859
MK-PUCIT [7]	2017	Urdu	652,852	3	18,150	18,728	7,602	44,480

4 WORD EMBEDDINGS FOR URDU

In this section, we present our second contribution, four different word embeddings for Urdu language. Word embeddings have provided a major breakthrough in deep learning techniques for various natural language processing tasks [28]. For generating word embeddings, words are mapped to vectors of real-numbers, such that words of similar meaning are mapped to similar vectors [29]. While creating embeddings, the context of a word is taken into consideration so that the vector representation captures both syntactic and semantic aspects of words. Hence, increasing the accuracy of finding the similarity and other relations among words [30].

There are three established techniques for generating word embeddings: Word2vec, Glove, and fastText [30–32]. Word2vec is supported by a leading enterprise, `Google`, and Glove is supported by a leading NLP research group at `Stanford University`. The libraries of these techniques can be used to generate customized word embeddings from a given data. FastText is the latest library for generating word embeddings which is developed by Facebook [32]. In addition to the fastText library, Facebook has also publicly released word embeddings for over 150 natural languages, including Urdu [33].

In this study, we have used the Facebook released word embeddings and created six additional word embeddings, by employing two state-of-the-art techniques on two corpora of Urdu text. That is, we have used MK-PUCIT corpus and another huge corpus of 28,006,880 tokens which is created from the raw text obtained from `NewsLink.pk`. For generating the word embeddings, the context window of size 5 and vectors of 300 dimensions are used. The detailed specifications of the text sources used for generating word embeddings are given in Table 5.

5 EXPERIMENTATION

For a thorough evaluation of the supervised learning techniques, we have performed experiments using two datasets, MK-PUCIT and IJNCLP, as IJCNLP is a publicly and freely available Urdu NER dataset. In particular, we used the two datasets to perform 10-folds cross-validation of 32 different experiments with two deep learning techniques (using seven types of word embeddings), and two traditional supervised learning techniques. The two deep learning techniques are Neural Networks

Table 5. Specifications of the Word Embeddings

Sr. No.	Name	Technique	Data Source	Training size	Vocabulary size
1	FBfT	fastText	Facebook	1,170,865	557,487
2	MKw2v	Word2vec	MK-PUCIT	926,776	40,231
3	NLw2v	Word2vec	Newslink	28,006,880	461,449
4	MKfT	fastText	MK-PUCIT	926,776	40,231
5	NLfT	fastText	Newslink	28,006,880	461,449
6	MKGlov	Glove	MK-PUCIT	926,776	40,231
7	NLGlov	Glove	Newslink	28,006,880	461,449

The first embeddings are obtained from Facebook thus their generation does not represent our contribution, whereas, the rest are our generated embeddings.

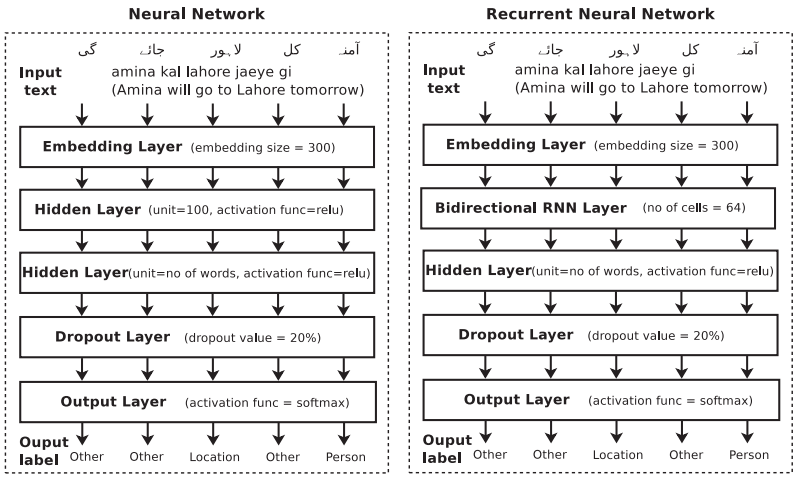


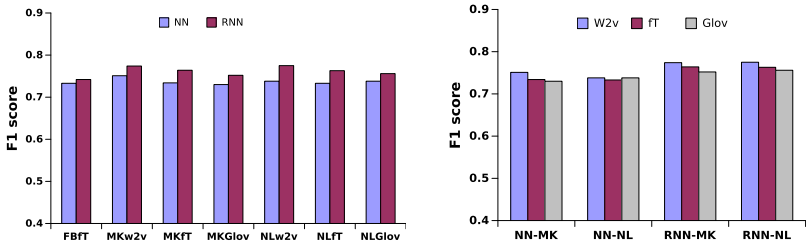
Fig. 1. Architectures of NN and ANN.

(NN) [34] and Recurrent Neural Networks (RNN) [35], whereas the traditional supervised learning techniques are Conditional Random Field (CRF) and Maximum Entropy Markov Model (MEMM).

For both, NN and RNN, a word-level embedding layer of 300 dimensions was used. Additionally, in NN, a window of size 5 was given as input to the embedding layer, its output was flattened, and fed to a dense layer of 100 units. In contrast, the input to RNN was a complete sentence. Subsequently, a bidirectional RNN layer composed of 64 cells was used. For both techniques, a dropout layer with 20% probability was used for regularization. Finally, *Relu* activation function was used in all hidden layers and *softmax* activation was used in the output layer. The detailed architectures of the two techniques are presented in Figure 1. For the traditional techniques, we used four features that play a pivotal role in deciding the NE type of a token. These features are, NE type of the previous token, two binary features indicating the start and end of a sentence, and context words feature that includes two preceding words and two subsequent words. We used training-testing ratio of 70:30, i.e., 70% of the tokens were randomly selected for training and the remaining 30% were used for testing. This may not lead to 70:30 ratio for each NE type, as the division is based on the total number of tokens rather than the quantity of NEs in each NE type. We have performed 10-folds cross-validations to reduce the bias that might have occurred during the random sampling of tokens. The experiments were carried out on a server with GPU-Nvidia Gforce TitanX having 12GB of memory. Finally, for each fold, Precision, Recall, and F1 scores were computed.

Table 6. Summary of the Results for MK-PUCIT and IJCNLP Corpora

Technique	Embeddings	MK-PUCIT dataset			IJCNLP dataset		
		P	R	F1	P	R	F1
MEMM	–	0.737	0.538	0.612	0.600	0.421	0.491
CRF	–	0.779	0.619	0.686	0.689	0.334	0.446
NN	FBfT	0.743	0.729	0.733	0.598	0.48	0.528
	MKW2v	0.756	0.748	0.751	0.571	0.486	0.519
	MKfT	0.748	0.724	0.734	0.585	0.475	0.519
	MKGlov	0.763	0.705	0.730	0.66	0.422	0.512
	NLW2v	0.764	0.732	0.738	0.615	0.477	0.533
	NLfT	0.747	0.722	0.733	0.615	0.461	0.522
	NLGlov	0.745	0.735	0.738	0.67	0.441	0.523
RNN	FBfT	0.742	0.743	0.742	0.681	0.519	0.587
	MKW2v	0.762	0.788	0.774	0.629	0.542	0.580
	MKfT	0.753	0.777	0.764	0.648	0.546	0.589
	MKGlov	0.741	0.765	0.752	0.638	0.527	0.575
	NLW2v	0.763	0.789	0.775	0.695	0.631	0.657
	NLfT	0.753	0.775	0.763	0.672	0.613	0.638
	NLGlov	0.745	0.764	0.756	0.663	0.56	0.604



(a) (b)

Fig. 2. Comparison of (a) NN and RNN, (b) Word2vec, fastText, and Glove.

6 RESULTS AND ANALYSIS

Table 6 presents the average Precision, Recall, and F1 scores of the 10-folds cross-validation. From the table, it can be observed that deep learning techniques outperformed the traditional supervised learning techniques for all the seven types of word embeddings and both corpora. Hence, establishing that the use of word embeddings in deep learning techniques outperforms the traditional supervised learning techniques. Furthermore, a performance comparison of the two deep learning techniques, NN and RNN, is presented in Figure 2(a). It can be observed from that figure that the F1 scores achieved by RNN are higher than NN for all types of word embeddings. It is because, NER is a sequence labeling problem and RNN is capable of capturing sequential data better than NN [36].

A comparison of three word embedding generation techniques, Word2vec, fastText, and Glove, is presented in Figure 2(b). From the figure, it can be observed that there is a slight difference between the performance of the three word embedding techniques. In particular, Word2vec performed better than fastText and Glove for both the deep learning techniques, hence, establishing its

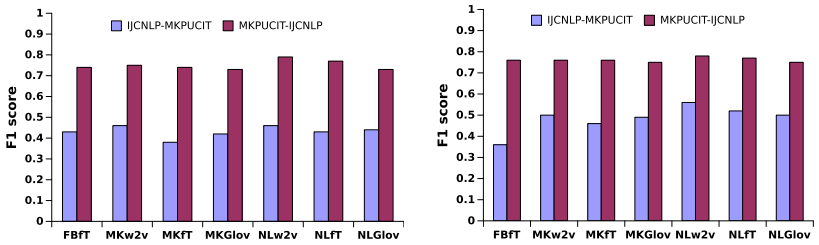


Fig. 3. Impact of training corpus size on performance: (a) NN and (b) RNN.

suitability for Urdu NER. Furthermore, it can be observed from Table 6 that the two types of word embeddings, that we have generated using fastText (namely, NKfT and NLfT), achieved higher F1 scores for RNN than the embeddings released by Facebook for both IJCNLP and MK-PUCIT datasets. A manual examination revealed that the reason for the higher performance of our embeddings stems from the quality of the raw text used for generating the embeddings. That is, a significant percentage of the vocabulary in the Facebook embeddings is non-Urdu, whereas we have used a cleaned text of Urdu language for generating embeddings.

It can be observed from Table 6 that the F1 scores achieved by all the techniques for the MK-PUCIT corpus are higher than the IJCNLP corpus. We hypothesize that it is due to the exceedingly large size of our MK-PUCIT corpus compared to the IJCNLP corpus, which offers better opportunities for learning and predicting NEs in Urdu text. To empirically test the hypothesis, we have performed experiments using IJCNLP and MK-PUCIT as training-testing pair and vice versa. These experiments are performed using all the seven word embeddings and both deep learning techniques. The results of the experiments shown in Figure 3 establish the usefulness of our MK-PUCIT corpus over IJCNLP corpus.

6.1 Error Analysis

To analyze the underlying reasons of misclassification by the techniques, while using MK-PUCIT corpus, we synthesized the results of the experiments. To this end, we identified all the 6,673 tokens in a testing dataset that were assigned an incorrect NE type by a majority of the supervised learning techniques. Our manual examination involving two researchers revealed that the five key reasons of the misclassification are: (a) incorrect tokenization, (b) scarcity of uncommon NEs in the training data, (c) the use of abbreviations or nicknames in the testing data set, (d) non-Urdu text in the testing data, and (e) incorrect disambiguation of the tokens that were candidates for multiple NE types. Below, we discuss each NE type separately and include examples of the misclassified NEs.

Misclassified Organizations. Among the 2,571 misclassified organizations, 29 were marked as PER, 231 as LOC, and the remaining 2,311 as O. For instance, سرینا انڈسٹریز (Serene industries) was tokenized into two separate tokens سرینا and انڈسٹریز which were marked as PER and O, respectively. Similarly, پنجاب یونیورسٹی (Punjab University) was incorrectly marked as پنجاب [LOC] and یونیورسٹی [O]. دارالعلوم‌شہابیہ (Darul-uloom-Shahabiya), and ڈبلیو ایچ او (WHO) are the examples of the 2311 NEs that were misclassified as O.

Misclassified Locations. Among the 1,976 misclassified locations, 65 were marked as PER, 124 as ORG, and the remaining 1,722 as O. The examples of the tokens that are misclassified as PER are: ہارون (Haroon), and اقبال (Iqbal), which are excerpts of ہارون آباد (Haroon abad), and اقبال پارک (Iqbal Park). Similarly, other examples of misclassified tokens are چیکوسلواکیہ (Czechoslovakia), and سعودیہ (Saudia instead of Saudi Arabia) as the former is a rarely found token in the text and the latter is an abbreviation.

Table 7. Misclassifications of the Three NE Types

	ORG	LOC	PER	O
ORG		231	29	2311
LOC	124		61	1722
PER	5	93		2097

Misclassified Persons. Among the 2,195 misclassified person names, 93 were marked as LOC, 5 as ORG, and 2,097 were classified as O. The examples of misclassified persons due to incorrect tokenization and scarcity of availability are: عارف عبدالمتین (Arif Abdul Mateen), and ظہیر کاشمیری (Zaeer Kashmiri).

The distribution of each misclassified NE type is also presented below in Table 7. It can be observed from the table that in majority of the cases (2,311 ORG, 1,722 LOC, and 2,097 PER) the techniques failed to recognized NEs and marked as O.

7 CONCLUSION

This article makes three key contributions for the advancement of the Urdu NER research. First, we have developed and publicly released the largest ever Urdu NER corpus (MK-PUCIT) by significantly extending our previously developed corpus. For the development of corpus, we defined two levels of annotation guidelines, illustrative examples, and the use of Kappa coefficient to minimize human bias. MK-PUCIT corpus contains 926,776 tokens and 99,718 NEs, which is a twofold increase from our previous corpus that contained 44,480 NEs. Furthermore, MK-PUCIT corpus is larger than majority of the well-established and widely used NER corpora for the Western languages. Hence, our released corpus provides the necessary basis for the future development and the evaluation of NER techniques, for an important resource-poor language.

Second, we have generated six types of word embeddings for Urdu language. These are the only publicly available embeddings for the Urdu language, beside the Facebook released embeddings. We have used two state-of-the-art techniques for generating word embeddings, fastText, Word2vec, and Glove, over two news corpora. We believe that our newly released diverse word embeddings will provide the community an opportunity for advancing Urdu NER research.

Third, we have evaluated the effectiveness of two deep learning techniques as well as two traditional supervised learning techniques, using the newly generated corpus and the word embeddings. This is the first ever attempt to thoroughly evaluate the performance of deep learning techniques using word embeddings in the context of Urdu NER. For the reliability of results, 10-folds cross-validation of experiments was performed. From the results of the experiments we conclude: (a) deep learning techniques outperform traditional supervised learning techniques, (b) RNN is more effective technique than NN, and (c) MK-PUCIT is a useful resource for Urdu NER.

The future work includes enhancing the accuracy of Urdu NER by improving the precision of tokenization and integrating POS information with deep learning. Furthermore, the trade-off between adapting the existing models for the eastern languages (e.g., Hindi, Persian, and Arabic) and developing custom models for the Urdu language warrants a thorough investigation.

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